UU-P-663B
August 24, 1972
SUPERSEDING
Fed. Spec. UU-P-663A
May 25, 1964

FEDERAL SPECIFICATION

PAPERBOARD, DRAWING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single and double faced paperboard in both rough and smooth surfaces for use in pencil, ink, and water color drawings.

1.2 Classification.

1.2.1 Types, grades, and classes. The paperboard shall be of the following types, grades, and classes, as specified (see 6.2).

Type I - Single faced.
Type II - Double faced.
Grade A - Drawing surface(s) 100 percent cotton or linen fibers.
Grade B - Drawing surface(s) 50 percent cotton or linen fibers.
Class 1 - Rough finish.
Class 2 - Smooth finish.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-I-528 - Ink, Drawing, Waterproof, Black.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:21Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 401 - Fiber Analysis of Paper and Paperboard.
T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 411 - Thickness (Caliper) of Paper and Paperboard.
T 459 - Wax Pick Test of Paper.
T 479 - Smoothness of Printing Paper.
RC-285- Sheffield Smoothness.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard sample. A standard sample shall be furnished with the invitation for bid purchase order (see 6.3) for comparison with the surface finish of class 1 items to be supplied. Class 1 paperboard shall conform to the requirements of this specification in all other respects.

3.2 Material.

3.2.1 Facing sheet. The fiber furnish of the facing sheet for grade A shall contain not less than 100 percent cotton or linen fiber. The facing sheet for grade B shall contain not less than 50 percent cotton or linen fiber and shall be free from unbleached or groundwood pulp.

3.2.2 Backing sheet. The backing sheet (back liner) of type I paperboard shall be of sufficient contrast to the facing sheet so as to establish the intended working surface. The backing sheet for type II shall conform to the applicable grade of facing sheet specified in 3.2.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:21Z
Check the source to verify that this is the current version before use.

3.2.3 Paper core. The paper core shall be made from cellulosic, fibrous material and may include reclaimed fiber.

3.3. Construction. The drawing paperboard shall be a combination board produced by pasting facing and backing sheets to the paperboard core. The finished drawing paperboard shall be flat and have even working surfaces free from mechanical defects. When tested as specified in 4.4.4, facing and type II backing sheets shall not separate or show separation from the paperboard core.

3.4 Dimensions. Drawing paperboard shall be furnished in the following sizes, as specified (see 6.2):

20 x 30 inches	22 x 30 inches
30 x 40 inches	40 x 60 inches

A tolerance of ± 1/8 inch will be allowed in each direction. The sheets shall be trimmed square on four sides with smooth, clean-cut edges. The finished paperboard shall be considered square if the difference between the two diagonals does not exceed 1/8 inch.

3.5 Physical requirements. The drawing paperboard shall conform to the requirements specified in table I. The thickness of the drawing paperboard shall be as specified (see 6.2).

TABLE I. Physical requirements

Characteristic	Facing sheet	Drawing paperboard
Basis weight, 17 x 22 - 500	Minimum, 32 pounds	
Thickness 1/		0.103 ± 0.008 inch 0.075 ± 0.005 inch 0.050 ± 0.005 inch
Finish (class 2) 2/ Bekk or heffield		Not less than 20 seconds Not more than 200 flow units.

1/ Thickness shall be as specified in the contract or order.
2/ Applicable to the drawing surface side of type I and both sides of type II paperboard.

3.5.1 Finish, class 1. Class 1 paperboard shall be equal to the standard sample specified in 3.1 with respect to surface finish.

3.6 Color. Unless otherwise specified, the facing sheets for types I and II, and the backing sheet for type II shall be white.

3.7 Identification and marking. Type I paperboard shall be identified by the manufacturer's name or trade mark on the back. Manufacturer's name or trade mark appearing as a colorless impression in one corner of all sizes of single or double faced paperboard is optional.

3.8 Surface bonding. The drawing surface(s) of each type, grade and class of paperboard shall have a critical wax strength number of 18A or higher. The surface(s) shall be free of lint, fuzz, or loose particles.

3.9 Drafting qualities.

3.9.1 Water-coloring and erasing (class 1 only). When tested as specified in 4.4.1, water colors shall flow freely and evenly over the surfaces of class 1 paperboard without any perceptible differences between the application to original surfaces or over erased areas.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:21Z
Check the source to verify that this is the current version before use.

3.9.2 Inking and erasing. Ink shall flow freely and evenly over the surface without showing any spreading, feathering, or ragged edges when inked as specified in 4.4.2. There shall be no appreciable spreading or feathering when inked after erasing as specified in 4.4.2.

3.9.3 Penciling and erasing. There shall be no visible change of the erased surfaces or perceptible linting or smudging when tested as specified in 4.4.3.

3.10 Workmanship. Drawing paperboard shall be flat and uniform in thickness throughout. Facing and backing sheets shall be free from hard spots, wrinkles, and shall be securely bonded to the paperboard core.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Sampling for inspection and test. Sampling for inspection and test shall be performed in accordance with MIL-STD-105, except where otherwise indicated. For sampling purposes, a lot shall consist of the same type, grade and class of drawing paperboard offered for inspection at one time.

4.2 Inspection of components and materials. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, examined and tested in accordance with this specification or referenced subsidiary specifications and standards.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. Examination of the end item shall be made for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth therein.

4.3.1.1 Examination of preparation for delivery. An examination shall be made to determine whether markings, materials, and workmanship comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed. The lot size shall be expressed in units of shipping containers. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 expressed in defects per hundred units. Defects shall be scored as listed below.

Examine	Defects
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence or method of application.
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps; loose or inadequate sealing, strapping, or stapling.
Contents	Number of intermediate packages per shipping container is less than specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:21Z
Check the source to verify that this is the current version before use.

4.3.1.1.1 Examination of the end item for average count. The sample unit for this examination is the package randomly selected from each shipping container selected in accordance with 4.3.1.1. The lot size shall be expressed in units of packages. The inspection level shall be S-2. The average number of drawing paperboards per package, of all units examined, shall be not less than the quantity specified.

4.3.1.2 Examination of the end item for visual defects. The drawing paperboard shall be examined for workmanship, color and finish. The lot size shall be expressed in units of sheets of drawing paperboards. The inspection level shall be S-3 and the AQL shall be 2.5 expressed in defects per hundred units. The sample unit shall be one sheet randomly selected from the packages selected in 4.3.1.1.1. No more than 2 sheets shall be selected from any one package. Both sides of the material shall be examined. Defects of each type shall be scored only once per sample unit as listed below.

Examine	Defect
Workmanship	Not clean: any mark or discoloration affecting appearance. Presence of dirt. Hole, tear, crease, wrinkle. Blister, lump or ply separation. Warp or buckle. Edges not clean cut; ragged, crushed, uneven. Manufacturers identification missing from back (type I only). Facing sheets not firmly attached at all points of contact.
Color and finish, as applicable	Not as specified. Does not compare favorably to bid sample. Not uniform. Backing of type I paperboard not in contrast to the facing sheet.

4.3.1.3 Examination of the end item for dimensional defects. The drawing aperboards shall be examined for dimensional defects. The lot size shall be expressed in units of sheets of drawing paperboard. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in defects per hundred units. The sample unit shall be one sheet selected from the packages selected in accordance with 4.3.1.1.1. Defects shall be examined as listed below.

Examine	Defect
Sheets: length or width	Varies from specified dimension by more than ± 1/8 inch.
Squareness	Difference between two opposite corner diagonals exceeds 1/8 inch.

4.3.2 Testing of the end item. The end item shall be tested for the applicable characteristics as specified in section 3 for each lot presented for examination for each type, grade and class of material. The tests shall be made in accordance with the indicated TAPPI method or as referenced in table II. The lot size shall be expressed in units of drawing paperboard of one type, grade and class. The sample unit shall be two sheets. No more than one sample unit shall be drawn from a package. Five sample units randomly selected throughout the lot shall be tested. Test results shall include all values on which the results are based and there shall be no failure to meet the lot average or sample unit requirements. All samples shall be conditioned and tested in accordance with the provisions of TAPPI T402.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:21Z
Check the source to verify that this is the current version before use.

TABLE II. Instructions for testing the end item

Characteristic	Specification reference		Requirements applicable to		Number of determinations per sample unit	Results reported as	
	Rqmt.	Test method or paragraph	Sample unit	Lot avg.		Pass or fail	Numerically to nearest
Atmospheric conditions	4.3.2	T402	All testing except drafting qualities.				
Fiber identification	3.2.1	T401			1 on composite 2/		5 percent
Basis weight	Table I	T410		X	1		1 pound
Thickness	Table I	T411		X	2		0.001 inch
Finish							
Bekk	Table I	T479		X	10		1 second
Sheffield	Table I	RC-285 1/		X	10		1 flow unit
Wax pick	3.8	T459	X		1	X	critical wax strength number
Ply separation	3.3	4.4.4	X		1	X	
Drafting qualities							
Water-coloring and erasing	3.9.1	4.4.1	X		1	X	
Inking and erasing	3.9.2	4.4.2	X		1	X	
Penciling and erasing	3.9.3	4.4.3	X		1	X	

1/ TAPPI Routine Control Method.
2/ The composite sample shall consist of approximately equal amounts drawn from all the sample units into a single sample.

4.4 Test methods.

4.4.1 Water-coloring and erasing. Rough finishes shall have water colors in washes of varying intensities applied by use of an artist's brush to the natural surface which shall contain areas that have been rubbed moderately with an electric erasing machine. The erasing machine, (speed 7500 revolution per minute (rpm)), shall be equipped with an eraser of medium abrasive texture. The surface shall then be carefully inspected for eveness of flow and change in appearance of erased areas.

4.4.2 Inking and erasing. Rough and smooth finishes shall have lines of varying width of black waterproof drawing ink conforming to the requirements of TT-I-528, drawn upon portions of the sample with a ruling pen and with free hand pen and carefully inspected for any spreading, feathering, or ragged edges visible to the naked eye. Allow a minimum of 5 minutes for the ink lines to dry and then erase them with an electric erasing machine (speed 7500 rpm) equipped with an eraser of medium abrasive texture. Using the erased surfaces repeat the drawing and erasing procedure until the lines have been drawn for a third time. Then examine the ink lines for spreading or feathering.

4.4.3 Penciling and erasing. Lines shall be drawn upon different portions of rough and smooth finish samples with a 2H pencil and erased with an electric erasing machine (speed 7500 rpm) equipped with an eraser of medium abrasive texture. The samples shall be inspected for any visible change of the erased surfaces and perceptible linting and smudging.

4.4.4 Ply separation. Condition two samples of drawing paperboard for 24 hours at 50 percent relative humidity and 73°F. After conditioning run the ball of the thumb over the edge at several different points on each sample. Evidence of ply separation shall indicate failure to meet the requirement of 3.3. Check the surfaces of the samples for the appearance of blisters which will also indicate failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:21Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Twenty-five sheets of drawing paperboard of one type, grade, class and size shall be completely wrapped in kraft paper having a minimum basis weight of 60 pounds per ream (24 x 36-500). The wrapping shall be secured with an adhesive or tape. The adhesive shall be applied so that contact of the adhesive is not made with the paperboard being packaged.

5.1.2 Level B. Twenty-five sheets of paperboard of one type, grade, class and size, only, shall be packaged with kraft paper specified in 5.1.1, in plastic or wax-coated kraft paper, or in a box conforming to PPP-B-636, class domestic.

5.1.3 Level C. Sheets of paperboard shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Fifty sheets of drawing paperboard packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, class V3c or V3s, style FTC except that the dimensional and weight limitations shall be waived. Each shipping container shall be closed, waterproofed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the box specification.

5.2.2 Level B. Fifty sheets of paperboard packaged as specified in 5.1 shall be packed in a box conforming to PPP-B-636, style RSC, FOL, or FTC, class domestic, except that the weight limitations shall be waived. Each shipping container shall be closed and securely sealed in accordance with the appendix of the box specification.

5.2.3 Level C. The drawing paperboard in quantities as specified (see 6.2), ackaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification and National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special markings required in the procurement documents, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use. The paperboard, drawing is used for water color, pencil and ink work in making posters, illustrations and displays. The rough finish (class 1) paperboard is generally for use with pencil and water colors. The smooth finish (class 2) paperboard is generally for use with pen and ink.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, class, thickness, and size required (see 1.2.1, 3.4 and 3.5).
(c) Color required when other than white (see 3.6).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Quantities required (see 5.2.3).
(f) Marking document required (see 5.3).

6.3 Standard sample. Standard comparison samples may be obtained upon pplication to the procuring agency issuing the invitation for bids.

Source: https://assist.dla.mil -- Downloaded: 20260-26T12:21Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army-GL
Navy-SA
Air Force-69

Review activities:

Army-GL, SM
Navy-SA
Air Force-69

Preparing activity:

GSA-FSS

User activities:

Army - MD
Navy - MC

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:21Z
Check the source to verify that this is the currentversion before use.